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Department of Computer Science and Engineering (AI&ML)

Project Title: - Hierarchical Position Encoding for Structure-Sensitive Semantic Dense Retrieval

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1. 1. Problem Definition

When the same sentence means opposite things depending on where it appears in data, can the embedding model tell the difference?

Conventional dense retrieval models primarily learn semantic similarity between text passages, but may fail to distinguish information whose meaning depends on hierarchical position, relational context, or its role within a larger knowledge structure. This can cause structurally distinct passages to receive similar embeddings, leading to incorrect retrieval in retrieval-augmented generation (RAG) systems.

HiSE (**Hierarchical Structure-aware Embeddings**) proposes a unified framework for integrating semantic meaning, document hierarchy, and relational knowledge into a shared retrieval system. It combines **Hierarchical Structural Addressing** (HiSA) to encode hierarchical paths into pre-attention embeddings and **Structure-Guided Attention Mechanism** (SGAM) to incorporate structural relationships into attention. Contrastive learning with structurally hard negatives will train the model to distinguish semantically similar but structurally different passages.

A central research objective is to investigate representation convergence and structural collisions: when and why distinct paths or relationships become indistinguishable in embedding space, and how path-based composition, relational rotations, and alternative geometric representations can preserve structural identity while retaining semantic similarity.

HiSE also investigates world-model construction through **pretrained representation knowledge**. By probing a pretrained embedding model for latent entity relationships, the framework aims to

construct a candidate knowledge graph that can guide structural learning and retrieval. This knowledge-transfer process will be integrated with query-side structural intent prediction, enabling queries to represent not only their semantic meaning but also their likely structural targets. A **POMDP**-inspired formulation will further investigate how structural information reduces uncertainty about a passage's position and relationships within a document.

The overall objective is to develop and evaluate a retrieval framework that jointly models semantic meaning, hierarchical structure, relational knowledge, and representation convergence, measuring whether these components improve retrieval accuracy, structural discrimination, and generalization across hierarchical documents and structured data.

2 Introduction

2.1 Background and Motivation

The global Retrieval-Augmented Generation (RAG) market was valued at $2.33 billion in 2025 and is projected to reach $81.51 billion by 2035, with a CAGR of 42.7% [1]. Document retrieval is the dominant function, accounting for 32.4% of all RAG revenue. Despite this rapid adoption, naive RAG systems achieve only 10–40% success rates in enterprise environments — with retrieval failure, not generation failure, identified as the primary bottleneck [2].

Over 80% of enterprise data is unstructured and hierarchically organized — financial reports, legal contracts, technical documentation, configuration files, and API specifications. Every query against these documents passes through an embedding model. The quality of that embedding model directly determines whether the correct context reaches the downstream language model.

2.2 The Core Problem

Every major embedding model in production — BGE-M3, Qwen3-Embedding, E5-Mistral, Nomic Embed — encodes token position as a single integer via Rotary Position Embedding (RoPE). RoPE computes attention scores as a function of relative sequence distance only:

$$\boldsymbol{Attention}(\boldsymbol{A}, \boldsymbol{B}) \sim \boldsymbol{f}(|\boldsymbol{pos_A} - \boldsymbol{pos_B}|)$$

This architecture is fundamentally blind to structural address — the depth in a document tree, the parent section identity, the sibling index — that governs what a token actually means in hierarchically structured documents.

The consequence is a systematic retrieval failure. The sentence "Revenue grew 40%" embedded under the heading "Q3 Performance > Revenue" and the same sentence under "Q3 Risks > Revenue" produce near-identical embedding vectors in every current production model (cosine similarity ≈ 0.97). A RAG system querying "what are the Q3 risks?" may retrieve the performance-framed chunk as equally or more relevant than the risk-framed chunk, causing the downstream language model to generate a confidently wrong answer. This is not a data problem or a scale problem — it is an inductive bias problem. No amount of additional training data corrects a mechanism that was never designed to encode structural position.

2.3 Research Gap

The benchmarks that all production embedding models optimize for — BEIR and MTEB — contain zero structurally-adversarial test pairs. If a benchmark never tests structural sensitivity, no model will optimize for it, and no researcher will detect the failure until it surfaces in a production system. This explains why the gap has persisted despite significant investment in embedding model research through 2023–2025. Existing work on structure-aware retrieval — KohakuRAG (2026), SPROUTRAG (EMNLP 2026), and "Equipping RAG with Document Structure Awareness" (2025) — addresses structural signals at the pipeline level: chunking strategy, indexing, or reranking. None modifies the positional encoding mechanism of the embedding model itself. The geometric representation of structural position in the embedding vector space remains an open problem.

2.4 Proposed Solution — HiSE

This project proposes HiSE (Hierarchical Structure-aware Embeddings), a fine-tuned retrieval embedding architecture incorporating three coordinated contributions.

HiSA (Hierarchical Structural Addressing) extends RoPE from one-dimensional sequence position to a four-dimensional structural address. The standard RoPE rotation angle for embedding dimension i is:

$$theta_i = m / 10000^{(2i / d)}$$

where m is the sequence position. HiSA replaces this with a direct-sum subspace decomposition across the full embedding space:

$$R^{768} = R_m^{384} + R_d^{154} + R_p^{154} + R_s^{76}$$

where each subspace receives an independent rotation encoding one structural dimension: m = sequence position (alpha = 0.5), d = depth in document hierarchy (beta = 0.2), p = parent path identity encoded via path-composed relational rotations (gamma = 0.2), s = sibling index (delta = 0.1). The structural rotation per axis uses the N-dimensional RoPE formulation R_x = exp(sum_i x_i * B_i) where B_i are mutually commuting generators, preserving the relative position property q_i^T * R(x_j - x_i) * k_j such that attention scores depend only on relative structural offset. The parent dimension p is encoded via path composition rather than hashing — eliminating collision and making the encoding algebraically invertible. Implementation uses low-rank structural projection at rank r = 64, achieving 144x FLOPs reduction over full-rank construction with negligible accuracy loss.

SGAM (Structure-Guided Attention Mask) adds a learned structural bias to attention scores before softmax during fine-tuning:

$$\boldsymbol{Attn(Q,K,V) = softmax((Q * K^T / sqrt(d_k)) + SGAM) * V}$$

where SGAM(i, j) = w_d * |d_i - d_j| + w_p * 1[p_i != p_j] + w_s * |s_i - s_j|, with w_d, w_p, w_s learned scalar weights. This propagates structural relationship signals into W_Q and W_K during fine-tuning — the model's learned weight matrices carry structural awareness permanently, not only when SGAM is applied.

Knowledge Transfer Pipeline extracts relational world structure from the frozen base model's weight space via learned probing layers, constructing a structural prior graph without requiring additional training data. This graph supervises fine-tuning as a secondary signal — the base model's own compressed world knowledge bootstraps the structural geometry of its fine-tuned successor.

HiSE is fine-tuned from bge-base-en-v1.5 using contrastive learning with structurally-adversarial hard negatives and curriculum-ordered training. The project also introduces **StructEval-Bench v1** — the first retrieval benchmark explicitly designed to measure structural sensitivity — operationalizing two new metrics: Structural Discrimination Score (SDS) and Retrieval Inversion Rate (RIR).

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3. Literature Review

Research on structure-aware representation has progressed through table encoders, HTML encoders, code structure encoders, and prompt-based approaches. However, these approaches do

not provide one architecture-level rotational encoding for hierarchical dense retrieval across multiple data formats.

Table 1: Literature review of structure-aware representation methods

Paper or System	Year	Contribution	Limitation
Attention Is All You Need	2017	Foundational transformer and sinusoidal positional encoding.	Models sequence position, not hierarchical document structure.
TUTA	2021	Tree-based coordinates for generally structured tables.	Focused on classification and table pre-training, not dense retrieval.
BEIR	2021	Heterogeneous zero-shot information retrieval benchmark.	Does not include structurally adversarial test pairs.
DOM-LM	2022	Depth, parent, and sibling information for HTML documents.	HTML-specific and additive; no unified retrieval architecture.
MTEB	2023	Large benchmark for text embedding models.	Does not directly expose structural discrimination failure.
Struc-EMB	2025	Structural context through LLM-based embedding prompts.	Prompt-level workaround; positional encoding remains unchanged.
Rethinking RoPE	2025	Blueprint for N-dimensional rotary position embedding.	Primarily discussed for vision and spatial domains.

3.1 Research Gap in Dense Retrieval

No existing production embedding model encodes a composite structural address across tabular, JSON, HTML/XML, Markdown, and complex document formats. Existing solutions address individual formats or use prompt concatenation, rather than changing the positional encoding layer of the attention mechanism.

3.2 Need for a New Benchmark

BEIR and MTEB evaluate broad retrieval quality but contain zero structurally adversarial pairs designed to test whether identical or similar text is distinguished by its location in a hierarchy.

StructEval-Bench v1 is proposed to fill this gap with 300-500 query-document pairs and explicit Structural Discrimination Score (SDS) and Retrieval Inversion Rate (RIR) metrics.

4 Motivation

The global Retrieval-Augmented Generation (RAG) market was valued at $2.33 billion in 2025 and is projected to reach $81.51 billion by 2035, with a CAGR of 42.7% [1]. Document retrieval is the dominant function, accounting for 32.4% of all RAG revenue. Despite this rapid adoption, naive RAG systems achieve only 10–40% success rates in enterprise environments — with retrieval failure, not generation failure, identified as the primary bottleneck [2].

Over 80% of enterprise data is unstructured and hierarchically organized — financial reports, legal contracts, technical documentation, configuration files, and API specifications. Every query against these documents passes through an embedding model. The quality of that embedding model directly determines whether the correct context reaches the downstream language model.

5. Objectives

- Design HiSA as a format-agnostic extension of RoPE using a composite structural address (m, d, p, s).
- Develop HiSE-Embed, a contrastive retrieval model with fewer than 300 million parameters.
- Construct StructEval-Bench v1 with 300-500 structurally adversarial query-document pairs across Markdown, tables, and JSON.
- Evaluate Structural Discrimination Score and Retrieval Inversion Rate against standard RoPE-based baselines.
- Maintain general retrieval quality on BEIR and MTEB subsets, with nDCG@10 within a two percent margin of the baseline.
- Use controlled ablation to isolate the effect of the positional encoding while keeping backbone, data, and compute constant.

- Release the model, benchmark, and HiSA implementation as reproducible open-source research artifacts.

6. Methodology Used

This project proposes HiSE (Hierarchical Structure-aware Embeddings), a fine-tuned retrieval embedding architecture incorporating three coordinated contributions.

HiSA — Hierarchical Structural Addressing

HiSA extends RoPE from one-dimensional sequence position to a four-dimensional structural address via direct-sum subspace decomposition of the full 768-dimensional embedding space:

$$R^{768} = R_m^{384}(direct\ sum)R_d^{154}(direct\ sum)R_p^{154}(direct\ sum)R_s^{76}$$

Subspace allocation: alpha = 0.50 for sequence position m, beta = 0.20 for depth d, gamma = 0.20 for parent path p, delta = 0.10 for sibling index s,

where alpha + beta + gamma + delta = 1.

Each subspace receives an independent rotation encoding one structural dimension. The rotation operator per axis follows the N-dimensional RoPE formulation:

$$R_x = \exp\left(sum_\{i = 1\}_i^{\{n\}x} * B_i \right)$$

where B_i are mutually commuting generators in the maximal abelian subalgebra of so(d), ensuring that rotations across subspaces do not interfere. The full HiSA rotation applied to a query vector q at structural address (m, d, p, s) is:

$$q_tilde = R_m(m) * q_m(direct\ sum)R_{struct(d,p,s)} * q_{struct}$$

where q_m is in R^384 and q_struct is in R^384, and R_struct is the composed structural rotation across the three structural subspaces.

The structural rotation R_struct is computed via path composition rather than hashing. For a node at depth D with sibling indices (s_1, s_2, ..., s_D) at each level of its ancestor path:

$$e_{node} = e_{root}(composed\ with)r_{child}^{s_1}(composed\ with)r_{child}^{s_2}(composed\ with)$$

$$\ldots(composed\ with)\ r_child\^{}(s_D)$$

where r_child is a learned unit complex vector in C^(d/2), and composition is element-wise complex multiplication. This guarantees: (1) collision-freeness — distinct paths produce distinct rotations; (2) algebraic invertibility — the parent embedding is recoverable as e_parent = e_node (composed with) conjugate(r_child); (3) structural smoothness — siblings differ by exactly one r_child step, cousins by two.

The key property preserved from standard RoPE is the relative position equivariance of attention scores:

$$q_{tilde_i}^T * k_{tilde_j} = q_i^T * R(x_j - x_i) * k_j$$

Attention scores depend only on the relative structural offset (x_j - x_i), not on absolute structural address. This means HiSE generalizes to documents with structural patterns not seen during training.

For compute efficiency, the structural rotation operates via low-rank projection:

$$R_{struct(x)} = U * R_{tilde(W_{down} * x_{struct})} * U^T$$

where W_down is in R^(r x d) with r = 64. FLOPs for the structural component drop by a factor of (768 / 64)^2 = 144 compared to full-rank construction. The structural signal requires only r = 64 dimensions because it encodes a bounded signal: approximately 10 possible depths, approximately 1000 parent identities, approximately 50 sibling positions.

Degenerate case guarantee: when d = 0, p = 0, s = 0 for all tokens, HiSA reduces exactly to standard RoPE. HiSE is a strict generalization of the baseline.

SGAM — Structure-Guided Attention Mask

SGAM adds a learned structural bias to the attention score matrix before softmax during fine-tuning.

Standard scaled dot-product attention:

$$Attn(Q,K,V) = softmax\left(Q * \frac{K^T}{sqrt(d_k)}\right) * V$$

HiSE attention with SGAM:

$$\boldsymbol{Attn(Q,K,V) = softmax\left(\left(Q * \frac{K^T}{sqrt(d_k)}\right) + SGAM\right) * V}$$

where the structural bias between token positions i and j is:

$$\boldsymbol{SGAM(i,j) = w_d * |d_i - d_j| + w_p * 1[p_i != p_j] + w_s * |s_i - s_j|}$$

w_d, w_p, w_s are learned scalar weights shared across all token pairs, optimized during fine-tuning. The indicator term 1[p_i != p_j] equals 1 when tokens i and j belong to different parent sections and 0 when they share a parent — applying a fixed penalty for cross-branch attention regardless of depth difference.

SGAM is active only during fine-tuning. Its function is to propagate structural relationship signals through backpropagation into W_Q and W_K. After fine-tuning, the learned weight matrices carry structural routing awareness permanently — SGAM is not required at inference.

The combined attention score between structurally-addressed tokens i and j is therefore:

$$\text{score}(i,j) = \frac{\tilde{q}_i^T \tilde{k}_j}{\sqrt{d_k}} + \text{SGAM}(i,j)$$

$$=\backslash frac\left\{q_{i_j^{\{T\}R(x_j - x_i)k}}\right\}\left\{\sqrt{\{d_k\}}\right\} + w_{d|d_i - d_j|} + w_p \backslash mathbf\{1\}[p_i \neq p_j] + w_{s|s_i - s_j|}$$

Two additive terms: the first is the HiSA relative structural rotation (geometric), the second is the SGAM structural distance bias (learned scalar). Both are differentiable. Both contribute to fine-tuning gradients.

Knowledge Transfer Pipeline

The knowledge transfer pipeline extracts relational world structure from the frozen base model's weight space without requiring additional training data. For every structural relation type r in {IS-

CHILD-OF, FOLLOWS, INTRODUCES, CONCLUDES, IS-SIBLING-OF}, a linear probing layer is trained on the frozen bge-base-en-v1.5 representations:

$$f(e_{head}, e_{tail}) = W_{probe} * [e_{head}; e_{tail}] \longrightarrow relation_{type}$$

where [e_head ; e_tail] is the concatenation of the two concept embeddings and W_probe is in R^(|R| x 2d). Training uses labeled structural relation pairs from Wikipedia infoboxes and document structure annotations. The probe is a single linear layer — it reads relational signal already present in the base model's geometry, not learned anew.

The extracted world graph G is then constructed by sampling concept pairs from the vocabulary and applying the probe:

for all (c_i, c_j) in sampled vocabulary pairs:

relation = argmax f(W_E[c_i], W_E[c_j])

if confidence > threshold:

add edge (c_i, relation, c_j) to G

G provides a structural prior over concept-level relationships — which concepts typically appear in introductions, which in conclusions, which in methodology sections. This prior supervises HiSE fine-tuning as a secondary signal alongside the contrastive loss, closing the loop: the base model's compressed world knowledge bootstraps the structural geometry of its fine-tuned successor with zero additional raw data.

Contrastive Fine-tuning

HiSE is fine-tuned using InfoNCE contrastive loss over structurally-constructed triplets:

$$L = -\log\left(\frac{\exp\left(\frac{sim(a,p)}{tau}\right)}{\left[\exp\left(\frac{sim(a,p)}{tau}\right) + sum_i \exp\left(\frac{sim(a,n_i)}{tau}\right)\right]}\right)$$

where a is the anchor chunk, p is a positive chunk from the same structural section, n_i are hard negative chunks selected to maximize text similarity while minimizing structural similarity — adjacent sections with near-identical surface text. tau = 0.07 is the temperature parameter.

Training follows a curriculum: easy negatives (different sections, different text) in early epochs, structurally-adversarial hard negatives (adjacent sections, near-identical text, different structural address) in later epochs.

The project also introduces StructEval-Bench v1 — the first retrieval benchmark explicitly designed to measure structural sensitivity — operationalizing two new metrics: Structural Discrimination Score (SDS) and Retrieval Inversion Rate (RIR).

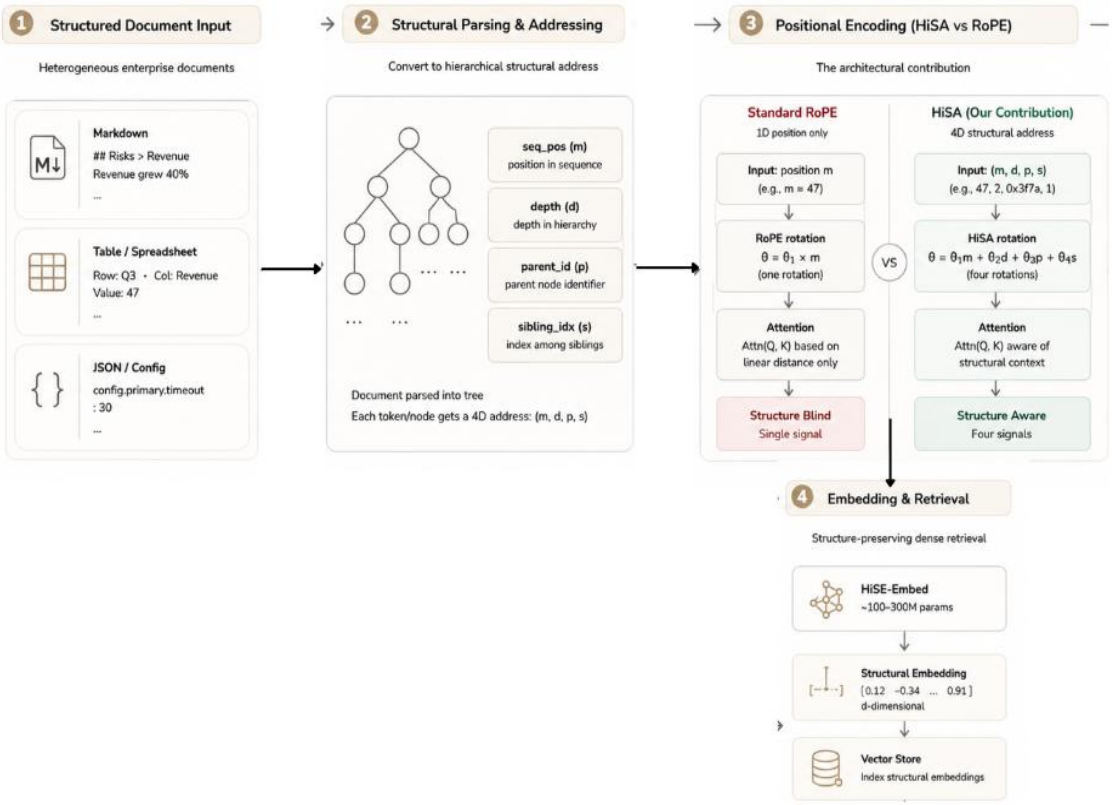


Fig 1: System Architecture

7. Development Plan

Table 2: HiSE development plan and success criteria

Phase	Timeline	Planned Work	Success Criterion
Failure Survey	Month 1-2	Measure the structural discrimination gap across baseline embedding models and formats.	Cosine similarity at least 0.90 on structurally opposed pairs; RIR baseline established.
HiSA Build	Month 2-3	Implement the four-dimensional address, parsers, and POMDP structural state estimator.	Zero-structure case matches standard RoPE.
Benchmark	Month 3-4	Build and document StructEval-Bench v1 with SDS and RIR metrics.	Benchmark released with annotation guidelines.
Training	Month 4-5	Train HiSE-Embed with structural hard negatives and run the ablation.	Statistically significant SDS improvement over the RoPE baseline.
Evaluation and Release	Month 6	Evaluate on BEIR/MTEB, release artifacts, and prepare the research submission.	nDCG@10 within two percent; submission by November 30, 2026.

7.1 Technology and Requirements

The implementation will use Python, PyTorch 2.x, CUDA, HuggingFace Transformers, Sentence-Transformers, FAISS, Python-Markdown, pandas, JSON parsing, and HTML/XML DOM parsing. BEIR, MTEB, and StructEval-Bench v1 will support evaluation. Docker will be used for reproducible development and training.

Training and prototype validation will use locally hosted NVIDIA GPU systems with at least 8 GB VRAM, 16-32 GB RAM, and approximately 500 GB SSD storage for checkpoints and embedding indexes.

8 Conclusion

5.1 Conclusion

This project addresses a fundamental and previously unmeasured architectural limitation in dense retrieval embedding models: the inability to encode hierarchical structural position. The contribution is not a domain-specific fix or a prompt-engineering workaround — it is a targeted modification to the positional encoding mechanism itself, grounded in formal mathematical properties and validated through controlled ablation and a purpose-built benchmark.

HiSE does not improve an existing model by training on more data. It corrects the inductive bias: replacing the assumption that sequential position is sufficient to represent position in a structured document. This is achieved through three coordinated mechanisms. HiSA extends RoPE to a multi-dimensional structural address via direct-sum subspace decomposition and path-composed relational rotations — encoding depth, parent identity, and sibling index as collision-free geometric priors in the embedding vector. SGAM propagates structural relationship signals into the model's learned weight matrices during fine-tuning, so structural awareness is not a post-hoc tag but a property of the model's internal representations. The knowledge transfer pipeline extracts relational world structure directly from the base model's frozen weight space, constructing a structural prior without requiring additional training data — the base model's own compressed knowledge bootstraps its structurally-aware successor.

The four research artifacts — HiSA (positional encoding mechanism), SGAM (structural attention mechanism), HiSE-Embed (fine-tuned model), and StructEval-Bench v1 (benchmark) — each constitute independent, verifiable contributions to the field of information retrieval and representation learning. Together they demonstrate a broader principle: vector spaces can encode world structure geometrically, not purely statistically, and this encoding is achievable through targeted architectural intervention rather than scale.

5.2 Future Scope

HiSA → Generative LLMs. The structural addressing mechanism is not retrieval-specific. Applied to generative model context windows, a structurally-aware LLM would attend differently to tokens in structurally-opposed positions during generation — distinguishing a claim in an introduction from the same claim restated in a conclusion. This would improve RAG answer quality beyond what better retrieval alone provides, addressing the full pipeline rather than the indexing layer.

World Graph Expansion. The knowledge transfer pipeline currently extracts structural relations from the embedding model's weight space via linear probing. Extending this to non-linear probing and multi-hop relation extraction would produce a richer world graph — one capable of encoding not just direct structural relations (IS-CHILD-OF, FOLLOWS) but compositional structural reasoning (this section introduces a concept that this later section refutes). This is the path toward a full geometric world model derived from existing pre-trained weights with zero additional data.

Cross-lingual Structural Retrieval. Extending HiSE to Indic languages where document structure is often the primary disambiguation signal — particularly in legal, administrative, and financial documents — is directly relevant to enterprise document search in the Indian market. The HiSA mechanism is language-agnostic; the required investment is structural parsing tooling for Indic document formats and a cross-lingual variant of StructEval-Bench.

StructEval-Bench v2. Expand the benchmark to legal contracts, clinical trial protocols, and financial filings with real-world enterprise annotation from domain experts. The v1 benchmark establishes the evaluation methodology; v2 establishes the scale and domain coverage required for adoption as a standard retrieval benchmark alongside BEIR and MTEB.

Query-Side Structural Routing. The BART-encoder structural intent classifier introduced in this work assigns approximate structural addresses to free-form queries at inference. A stronger version — trained on large-scale (query, relevant-chunk-structural-address) pairs with reinforcement from retrieval feedback — would make the query-side routing a first-class contribution rather than an auxiliary component, enabling fully symmetric structural alignment between query and document embeddings.

9. References

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